

SQL Business Insights from Walmart Sales Data

Key Business Questions & Insights

This section details the key business questions analyzed using SQL queries. Each query helps uncover insights into Walmart's sales trends, customer behavior, and financial performance.

1. Payment Methods and Transaction Volume

 **Question:** What are the different payment methods, and how many transactions and items were sold with each method?

 **Purpose:** Understanding the distribution of payment methods helps in optimizing checkout processes and assessing customer payment preferences.

2. Best-Rated Category in Each Branch

 **Question:** Which category received the highest average rating in each branch?

 **Purpose:** Identifying the best-rated categories allows Walmart to highlight top-performing products and enhance customer satisfaction strategies.

3. Peak Sales Days per Branch

 **Question:** What is the busiest day of the week for each branch based on transaction volume?

 **Purpose:** Helps in workforce planning and inventory management by identifying the busiest sales days.

4. Sales Volume by Payment Method

 **Question:** How many items were sold through each payment method?

 **Purpose:** Evaluates customer purchasing behavior and the efficiency of different payment methods.

5. Ratings Analysis by Category and City

 **Question:** What are the average, minimum, and maximum ratings for each category in each city?

 **Purpose:** Provides insights into customer satisfaction across different locations and product lines.

6. Profitability Analysis by Category

 **Question:** What is the total profit for each category, ranked from highest to lowest?

 **Purpose:** Determines the most profitable product categories, assisting in inventory and pricing strategies.

7. Most Popular Payment Methods per Branch

 **Question:** What is the most frequently used payment method in each branch?

 **Purpose:** Identifies preferred payment options, allowing for better payment processing strategies.

8. Transactions by Time Shift

 **Question:** How many transactions occur in each shift (Morning, Afternoon, Evening) across branches?

 **Purpose:** Helps in staffing decisions and optimizing store operations for different time slots.

9. Revenue Decline Analysis

 **Question:** Which branches experienced the largest decrease in revenue compared to the previous year?

 **Purpose:** Identifies branches with declining revenue, helping Walmart take corrective measures.

10. Total Sales per Category

 **Question:** What is the total sales per category?

 **Purpose:** Determines the revenue contribution of different product categories.

11. Average Spending per Transaction by Payment Method

 **Question:** What is the average spending per transaction for each payment method?

 **Purpose:** Analyzes how much customers spend based on their preferred payment method.

12. Branch with the Highest Average Rating

 **Question:** Which branch has the highest average rating?

 **Purpose:** Recognizes top-performing branches based on customer feedback.

13. Best-Selling Categories per City

 **Question:** What are the top 2 best-selling product categories in each city?

 **Purpose:** Identifies the most popular product categories for regional marketing strategies.

14. Highest Revenue Month

 **Question:** Identify the month with the highest revenue overall.

 **Purpose:** Helps in understanding seasonal trends and preparing for peak sales periods.

15. City-Wise Sales Performance per Category

 **Question:** Which city had the highest total sales in each category?

 **Purpose:** Determines city-wise demand variations for better inventory allocation.

16. Product Line Contribution to Total Revenue

 **Question:** Find the percentage contribution of each product line to the total revenue.

 **Purpose:** Evaluates how each product line contributes to overall sales, helping in decision-making for product expansion or discontinuation.